

1 Methods

1.1 Fine-scale H&E–MALDI-MSI alignment

We perform fine-scale alignment of MALDI-MSI measurements to H&E histology after an initial fiducial-based registration. Let $\mathcal{N} = \{1, \dots, n\}$ be the set of MSI spots, and let

$$\Omega_{\text{MSI}} = \{u_1, \dots, u_n\} \subset \mathbb{R}^2$$

denote their positions on the slide. Similarly, let $\mathcal{M} = \{1, \dots, m\}$ be the set of H&E pixels and

$$\Omega_{\text{HE}} = \{v_1, \dots, v_m\} \subset \mathbb{R}^2$$

denote their positions on the H&E image domain. The data is given by a preprocessed spots times MSI features matrix

$$X \in \mathbb{R}^{n \times p}$$

such that each feature has mean zero and variance one, and low variance / low quality features have been removed. We assume that an initial transformation

$$T_0 : \Omega_{\text{MSI}} \rightarrow \mathbb{R}^2$$

maps MSI coordinates into the H&E coordinate system with an accuracy of a few MSI pixels. Fine alignment aims to estimate the remaining local deformation between both modalities.

As an anatomical reference signal, we extract a binary mask

$$S_{\text{HE}} : \Omega_{\text{HE}} \rightarrow \{0, 1\}$$

from the H&E image. The mask represents a spatially informative tissue structure expected to have corresponding molecular signatures in the MSI data. In liver tissue, we use a sinusoid mask derived from H&E colour information, as described in Supplementary Section 2.1. Since the native H&E resolution is substantially higher than the MSI resolution, the binary mask is converted into a continuous field

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

by locally normalised Wendland kernel smoothing. Details of the kernel, boundary correction and interpolation are given in Section 2.2.

For any transformation T , evaluation of f at the mapped MSI positions yields a soft mask $s_T = (s_{T,i})$, where

$$s_{T,i} = f(T(u_i)) , \quad i \in \mathcal{N}.$$

The soft mask s_{T_0} becomes our registration target that guides the learning of a corresponding, linear MSI metafeature. For this purpose, we perform canonical correlation analysis of X with s_T using a robust, penalized version of canonical correlation analysis [1]. It guards against picking spurious MSI features and ultimately amounts to solving a ridge regression problem, see Supplementary Methods 1.1.

Further, because the initial transformation T_0 is expected to be most reliable near the fiducials and less reliable far away from any fiducial marker, metafeature learning is restricted to peri-fiducial spots in a certain radius of the fiducials. The radius of the peri-fiducial regions is selected by repeated, spatially balanced fiducial-level cross-validation, see Supplementary Methods 2.3. After selecting the peri-fiducial radius, the metafeature

$\hat{w}^{(0)} \in \mathbb{R}^p$ is estimated using all fiducials and all MSI positions within the selected radius. This gives the initial full-slide MSI pattern

$$q = X\hat{w}^{(0)},$$

The pattern q is the target of the geometric alignment step and kept fixed throughout optimisation.

Residual geometric correction is performed in two stages. We first fit a globally smooth thin-plate spline displacement field,

$$T_1(u_i) = T_0(u_i) + \Delta^{\text{TPS}}(u_i),$$

by maximising the correlation between the fixed MSI metafeature pattern $q^{(0)}$ and the transformed H&E target s_{T_1} , subject to the classical TPS bending penalty. This is the principal fine-alignment step. It corrects broad residual deformation while retaining a smooth and readily interpretable displacement field. Details of the TPS parameterisation, objective and regularisation are given in Supplementary Section 2.7.

Tissue preparation may introduce local tears for which a globally smooth transformation is inappropriate. We therefore apply an optional tear-aware correction after fitting T_1 . Candidate tears are identified by combining two independent criteria: unusually rapid spatial changes in the TPS displacement field and bright, weakly stained regions in the H&E image. The latter criterion is used only as a permissive anatomical gate, because physiological structures such as vessels and sinusoids may also appear bright.

The detected tear mask is morphologically closed, skeletonised and pruned to obtain a one-dimensional representation of the tear. Graph edges crossing this skeleton are removed from the MSI neighbourhood graph, thereby permitting relative movement across the tear while retaining smoothness within either side. MSI spots are partitioned into three disjoint sets,

$$\mathcal{N} = \mathcal{G} \dot{\cup} \mathcal{U} \dot{\cup} \mathcal{C},$$

where $\mathcal{G}$ contains trusted spots that remain fixed after the TPS alignment, $\mathcal{U}$ contains non-core spots subjected to local tear-aware refinement, and $\mathcal{C}$ denotes the tear core.

The residual displacement of anchored components of $\mathcal{U}$ is estimated by maximising fixed-standardisation agreement between $q^{(0)}$ and the transformed H&E target, while imposing a squared graph-Laplacian penalty on the cut graph. The regularisation strength is calibrated using pseudo-movable regions sampled from trusted tissue. Spots in $\mathcal{C}$ do not contribute to the registration objective and therefore cannot distort the alignment of surrounding intact tissue. Their coordinates are estimated afterwards by harmonic extension from the appropriate side of the tear. Full details are given in Supplementary Section 2.7.

The procedure assigns each MSI spot one of four status labels {trusted, locally adjusted, tear-core extrapolated, unresolved}. The trusted set can be used without relying on the tear-aware correction. The larger corrected output additionally contains locally adjusted and resolved tear-core spots, while retaining the status labels for downstream filtering.

After estimating the final transformation $\hat{T}$, the MSI metafeature $\hat{w}^{\text{final}}$ is re-estimated using the smooth target $s_{\hat{T}}$ on all resolved non-core spots, with the same standardised MSI matrix and ridge penalty. Stability of the learned spatial pattern is summarised by the correlation of $X\hat{w}^{(0)}$ and $X\hat{w}^{(\text{final})}$. This score-level comparison is preferred over direct comparison of the metafeature vectors because correlated MSI features may yield different coefficient vectors representing nearly identical spatial patterns.

2 Supplementary Methods

2.1 Generation of the H&E-derived anatomical mask

Let

$$M : \Omega_{\text{HE}} \rightarrow \{0, 1\}$$

denote the valid-pixel mask of the H&E image taking value 1 for all pixels considered foreground/tissue pixels to be analysed.

In liver tissue, sinusoids provide an informative anatomical marker for fine-scale alignment of H&E and MSI images. In two-dimensional tissue sections, they form bright, elongated structures with a characteristic spatial organisation at the scale of a few cells. To identify sinusoids, the H&E image is converted to HSV colour coordinates. A permissive empirical threshold on brightness and hue is used to identify candidate sinusoid pixels. Exterior background is removed, as are pixels belonging to vessels identified using Cellpose or an equivalent validated segmentation procedure. The result is another (possibly soft) mask

$$S_{\text{HE}} : \Omega_{\text{HE}} \rightarrow \mathbb{R}$$

In the case of the sinusoids, S_{HE} also assumes binary values $\in \{0, 1\}$. For tissues in which a sinusoid mask is not appropriate, the same alignment framework can be applied to another one-dimensional H&E-derived marker, such as a vessel mask, nuclei-density map, image gradient, edge map, blob-detector response or a selected component from a PCA or non-negative matrix factorisation of the H&E image.

2.2 Wendland smoothing and interpolation of the H&E mask

To evaluate whether a transformed MSI spot (whose position typically does not coincide with a H&E pixel position) lies inside the registration target (the sinusoids), one has to extend the alignment mask S_{HE} to a smooth map on the entire real plane. This is done by convolving it with a compactly supported Wendland kernel. For support radius ρ , this kernel is defined as

$$K_{\rho}(x) = \frac{1}{C_{\rho}} \left(1 - \frac{\|x\|_2^2}{\rho^2} \right)_+^4 \left(4 \frac{\|x\|_2^2}{\rho^2} + 1 \right),$$

where $(a)_+ = \max(a, 0)$ denotes the positive part of a , and C_{ρ} is a normalising constant such that $\int_{\mathbb{R}^2} K_{\rho}(x) dx = 1$. The support radius ρ of the kernel is chosen as the minimum radius such that approximately all H&E pixels are covered by discs around the mapped MSI spots. It is set to

$$\rho = \frac{\sqrt{2}}{2} \cdot \text{median}_{(i,j)} \|T_0(u_i) - T_0(u_j)\|_2,$$

where (i, j) runs through all pairs of direct horizontal or vertical neighbours on the original rectangular MSI grid.

We extend both binary masks M and S_{HE} to a joint (infinite) rectangular grid $\mathcal{G}$ by padding missing values with zeros. Ordinary zero-padded convolution $S_{\text{HE}} * K_{\rho}$ on this grid would incorrectly bias regions at the borders of the tissue towards zero. We therefore normalize for local tissue density, given by $M * K_{\rho}$. Our smoothed mask $f : \mathcal{G} \rightarrow \mathbb{R}$ is a Naradaya-Watson estimator

$$f(z) = \frac{(S_{\text{HE}} * K_{\rho})(z)}{(M * K_{\rho})(z)}$$

for all positions z for which the denominator in this expression is nonzero and $f(z) = 0$ otherwise. The domain of f is extended to $\mathbb{R}^2$ by bilinear interpolation.

For a given Transformation T , denote by

$$s_T = (f(T(u_i)))_{i=1,\dots,n}$$

the evaluation of the smooth mask at the transformed MSI coordinates.

2.3 MSI metafeature learning on a given set of MSI spots

The initial transformation T_0 gives rise to the anatomical registration target s_{T_0} . We search a MSI metafeature (a linear combination of basic MSI features) whose pattern agrees well with s_{T_0} . As the initial transformation T_0 is expected to be most accurate near the fiducials, the MSI metafeature is therefore learned from peri-fiducial MSI positions only rather than using all MSI positions.

For an arbitrary index set $\mathcal{I} \subseteq \{1, \dots, n\}$ of MSI spots, denote by $s_{T_0, \mathcal{I}}$ and $X_{\mathcal{I}}$ the sub-vector respectively sub-matrix obtained by restricting the MSI indices to the set $\mathcal{I}$. We estimate the ridge coefficients and an unpenalised intercept as

$$(\hat{\alpha}_{\mathcal{I}}, \hat{\beta}_{\mathcal{I}}) = \arg \min_{\alpha, \beta} \left\{ \frac{1}{|\mathcal{I}|} \|s_{T_0, \mathcal{I}} - \alpha \mathbf{1} - X_{\mathcal{I}} \beta\|_2^2 + \lambda \|\beta\|_2^2 \right\}, \quad (1)$$

using the fixed penalization $\lambda = 0.05$. The intercept accounts for the mean anatomical target and for possible differences between the feature means in $\mathcal{I}$ and those on the complete slide. The corresponding metafeature direction is the normalized coefficients vector

$$\hat{w}_{\mathcal{I}} = \frac{\hat{\beta}_{\mathcal{I}}}{\|\hat{\beta}_{\mathcal{I}}\|_2}. \quad (2)$$

Ridge regression is equivalent to maximising a regularised canonical correlation between the linear MSI score $\tilde{X}_{\mathcal{I}} w$ and the anatomical target $s_{T_0, \mathcal{I}}$ [1]. In this formulation, the empirical variance of the candidate MSI score is augmented by $\lambda \|w\|_2^2$. The regularisation therefore suppresses unstable directions with small empirical variance while retaining the objective of finding the MSI pattern most strongly correlated with the anatomical target.

Under the adopted full-slide feature standardisation, $\lambda = 0.05$ corresponds approximately to adding 0.05 to the diagonal of the empirical feature covariance matrix. This is small relative to the marginal variance of a standardised feature, while preventing unstable amplification of poorly supported feature directions.

The sign of $\hat{w}_{\mathcal{I}}$ is chosen such that

$$\rho(\tilde{X}_{\mathcal{I}} \hat{w}_{\mathcal{I}}, s_{T_0, \mathcal{I}}) \geq 0.$$

2.4 Construction of the fiducial neighbourhoods

It remains to find a suitable set of MSI pixels $\mathcal{I}$ for which the initial registration T_0 is good enough, i.e., for which $s_{T_0, \mathcal{I}}$ can have a correlating MSI metafeature. As registration should be best near the fiducial markers, $\mathcal{I}$ will be the union of discs around the fiducial markers with a certain radius. Let $h_1, \dots, h_K \in \Omega_{\text{HE}}$ denote the H&E fiducial coordinates. Each MSI position is assigned to its nearest fiducial,

$$k_i = \arg \min_{1 \leq k \leq K} \|T_0(u_i) - h_k\|_2,$$

and its distance to that fiducial is $d_i = \|T_0(u_{k_i}) - h_k\|_2$. Ties are resolved deterministically. This ownership assignment is fixed for all candidate radii and ensures that training and validation sets remain disjoint even when peri-fiducial discs overlap.

For fiducial k , let

$$m_k = |\{i \mid k_i = k\}|$$

denote the size of its complete nearest-fiducial territory.

Next, We construct $B = 10$ repeated bipartitions

$$A_b \dot{\cup} B_b = \{1, \dots, K\}, \quad b = 1, \dots, B,$$

of the fiducials that will serve as the centers of training and test regions. Each split is chosen to balance the number of assigned MSI spots while distributing both fiducial sets across the slide. The split construction does not depend on a candidate radius, so that the same fiducial split is retained for all radii during cross-validation. The resulting split is used in both directions for training and validation later.

For each split, one fiducial is first selected uniformly at random and assigned to A_b , and a second distinct fiducial is selected uniformly and assigned to B_b . Let

$$m_A = \sum_{k \in A_b} m_k, \quad m_B = \sum_{k \in B_b} m_k$$

denote the current territory sizes. While unassigned fiducials remain, the next fiducial is added to the set with the smaller current total, with ties assigned to A_b . To favour spatial coverage, the fiducial is sampled according to the k -means++ rule [2]. Specifically, if $\mathcal{K}_{\text{rem}}$ denotes the unassigned fiducials and $C \in \{A_b, B_b\}$ is the set currently being expanded, fiducial $k \in \mathcal{K}_{\text{rem}}$ is selected with probability proportional to

$$w_C(k) = \min_{j \in C} \|h_k - h_j\|_2^2.$$

The selected fiducial is then added to C , and the corresponding total S_C is updated by m_k .

For a fiducial set $A \subseteq \{1, \dots, K\}$ and radius r , define

$$\mathcal{I}_A(r) = \{i : k(i) \in A, \quad d_i \leq r\}, \quad n_A(r) = |\mathcal{I}_A(r)|.$$

For a subset of fiducials $A \subseteq \{1, \dots, K\}$, define the target variance within the corresponding peri-fiducial region as

$$v_A(r) = \text{Var} \{s_{T_0}(i) : i \in \mathcal{I}_A(r)\}.$$

and complete-slide target variance as $v_{\text{all}} = \text{Var} \{s_{T_0}(i) : i \in \mathcal{N}\}$.

The minimum candidate radius $r_{\min}$ is the smallest radius for which, for both training halves of every cross-validation split,

$$n_A(r) \geq 2p \quad \text{and} \quad v_A(r) \geq \frac{1}{2} v_{\text{all}}.$$

Recall that p is the number of retained MSI features. The first condition requires a training sample appreciably larger than the feature dimension. The second excludes very small peri-fiducial regions in which the anatomical target has insufficient spatial variation to identify a correlated MSI pattern. These conditions are used only to define the lower end of the radius search.

The smallest radius covering all valid MSI positions is

$$r_{\max} = \max_i d_i.$$

The radii candidate set contains $r_{\min}$, $r_{\max}$, and 8 further equally spaced intermediate empirical quantiles of the values d_i lying between these two radii.

2.5 Constnution of the reliable MSI spot set by split-set validation

For candidate radius r and split b , fit $\hat{w}_{A_b}(r)$ using $\mathcal{I}_{A_b}(r)$. Evaluate the resulting MSI pattern on the complementary fiducial territories,

$$C_{A \rightarrow B}^{(b)}(r) = \rho \left(\tilde{X}_{\mathcal{I}_{B_b}(r)} \hat{w}_{A_b}(r), s_{T_0, \mathcal{I}_{B_b}(r)} \right).$$

Reversal of the roles of the two halves A_b and B_b yields $C_{B \rightarrow A}^{(b)}(r)$. The two correlations $C_{A \rightarrow B}^{(b)}(r)$ and $C_{B \rightarrow A}^{(b)}(r)$ are combined on the Fisher scale,

$$Z_b(r) = \frac{1}{2} \left\{ \text{atanh} \left[C_{A \rightarrow B}^{(b)}(r) \right] + \text{atanh} \left[C_{B \rightarrow A}^{(b)}(r) \right] \right\},$$

and the radius-specific estimate of the correlation score is

$$C(r) = \tanh \left(\text{median}_{1 \leq b \leq B} Z_b(r) \right)$$

This criterion evaluates whether a metafeature learned near one set of fiducials predicts the smooth anatomical target in independent fiducial territories. It therefore assesses the stability of the resulting spatial pattern $\tilde{X}w$, rather than the coefficient vector w itself.

Let $C_{\max} = \max_r C(r)$. The selected radius is the largest candidate satisfying $C(r) \geq C_{\max} - 0.01$. This favours the use of more tissue whenever its held-out performance is practically indistinguishable from the best observed performance.

After selecting $\hat{r}$, the initial metafeature $\hat{w}^{(0)}$ is chosen according to Equations (1),(2), letting $\mathcal{I} = \mathcal{I}_{\hat{r}}$. Its sign is chosen such that $X_{\mathcal{I}} \hat{w}^{(0)}$ has a positive correlatoin with the target mask $s_{T_0, \mathcal{I}_{\hat{r}}}$. The full-slide metafeature used for alignment is

$$q^{(0)} = X \hat{w}^{(0)}.$$

2.6 Smooth TPS alignment

Starting from the initial fiducial-based transformation T_0 , we estimate a globally smooth residual correction,

$$T_1(u) = T_0(u) + \Delta^{\text{TPS}}(u).$$

This constitutes the principal fine-scale alignment step. The TPS field is intended to capture broad, smoothly varying residual deformation; more local motion associated with tissue tears is treated separately in the subsequent tear-aware refinement.

Low-rank TPS representation. Let

$$c_1, \dots, c_K \in \mathbb{R}^2$$

denote the TPS centres. To keep the model computationally feasible for slides containing several hundred thousand MSI spots, we use a low-rank TPS representation with approximately

$$K_0 = 144$$

centres in total, rather than placing one centre at each MSI spot.

Let W and H denote the width and height of the bounding box of the valid MSI positions, measured in MSI-pixel units. The numbers of grid positions in the two coordinate directions are chosen as

$$K_x = \max \left\{ 4, \text{round} \left(\sqrt{K_0 \frac{W}{H}} \right) \right\}$$

and

$$K_y = \max \left\{ 4, \text{round} \left(\sqrt{K_0 \frac{H}{W}} \right) \right\}.$$

Thus,

$$K = K_x K_y \approx K_0,$$

while the grid approximately follows the aspect ratio of the measurement region. The centres are placed on a regular Cartesian grid covering the bounding box of the valid MSI positions. This regular, low-resolution placement is deliberately chosen to restrict the TPS to broad residual deformation and to avoid adapting the basis to local irregularities of the tissue mask.

For displacement coordinate $\ell \in \{1, 2\}$, the residual field is represented as

$$\Delta_\ell^{\text{TPS}}(u) = \sum_{k=1}^K \theta_{k\ell} \phi(\|u - c_k\|_2),$$

where

$$\phi(r) = r^2 \log \left(\frac{r}{r_0} \right), \quad \phi(0) = 0,$$

and r_0 is fixed to one MSI-pixel spacing. The particular value of r_0 affects only components belonging to the affine null space of the TPS penalty.

No additional affine term is included in Δ^{TPS} . Translation, rotation, global scaling and shear are assumed to have been accounted for by the preceding fiducial-based transformation T_0 . Consequently, the TPS correction is restricted to non-affine residual deformation. To remove the affine null space from the radial representation, the coefficients satisfy the standard side constraints

$$\sum_{k=1}^K \theta_{k\ell} = 0, \quad \sum_{k=1}^K \theta_{k\ell} c_{k,1} = 0, \quad \sum_{k=1}^K \theta_{k\ell} c_{k,2} = 0, \quad \ell \in \{1, 2\}.$$

TPS bending energy. The classical TPS bending energy measures the integrated spatial curvature of the displacement field. For the two-dimensional displacement

$$\Delta^{\text{TPS}} = (\Delta_1^{\text{TPS}}, \Delta_2^{\text{TPS}})^\top,$$

it is defined as

$$E_{\text{bend}}(\Delta^{\text{TPS}}) = \sum_{\ell=1}^2 \int_{\mathbb{R}^2} \left[\left(\frac{\partial^2 \Delta_\ell^{\text{TPS}}}{\partial u_1^2} \right)^2 + 2 \left(\frac{\partial^2 \Delta_\ell^{\text{TPS}}}{\partial u_1 \partial u_2} \right)^2 + \left(\frac{\partial^2 \Delta_\ell^{\text{TPS}}}{\partial u_2^2} \right)^2 \right] du.$$

The energy is small for slowly varying displacement fields and increases when the deformation contains strong local curvature. Its null space consists of affine functions; these have zero second derivatives and therefore zero bending energy. In the present model, this null space is excluded by omitting the affine TPS component and imposing the side constraints above.

Let

$$\Theta = (\theta_{k\ell})_{k=1, \dots, K; \ell=1, 2} \in \mathbb{R}^{K \times 2}$$

and define the TPS kernel matrix

$$K_{\text{TPS},jk} = \phi(\|c_j - c_k\|_2).$$

Under the side constraints, the bending energy can be evaluated, up to the conventional constant associated with the definition of ϕ , as

$$E_{\text{bend}}(\Delta^{\text{TPS}}) = \text{tr}(\Theta^\top K_{\text{TPS}} \Theta).$$

Parameter estimation. The TPS coefficients are estimated by maximising agreement between the fixed MSI metafeature pattern $q^{(0)}$ and the transformed smooth H&E target, while penalising deformation curvature:

$$\hat{\Delta}^{\text{TPS}} = \arg \max_{\Delta^{\text{TPS}}} \left[\rho(q^{(0)}, s_{T_1}) - \lambda_{\text{TPS}} E_{\text{bend}}(\Delta^{\text{TPS}}) \right],$$

subject to the TPS side constraints. Here,

$$s_{T_1}(i) = f(T_0(u_i) + \Delta^{\text{TPS}}(u_i)).$$

The correlation term rewards agreement of the spatial MSI and H&E patterns, whereas the bending-energy term prevents the TPS from explaining local cross-modal disagreement through unnecessarily curved deformations.

With approximately $K_0 = 144$ centres, the TPS kernel matrix has only about 144×144 entries. Evaluation of the TPS field at the MSI positions nevertheless involves an $n \times K$ basis matrix. For large slides, this matrix is evaluated in blocks during optimisation rather than stored in full. This preserves the low-rank TPS model while making the calculation feasible for approximately 5×10^5 MSI spots.

2.7 Tear detection and tear-aware refinement

TPS displacement-gradient score. Let

$$\Delta_i^{\text{TPS}} = \Delta^{\text{TPS}}(u_i).$$

On the eight-neighbour MSI graph, define

$$g_{ij} = \frac{\|\Delta_i^{\text{TPS}} - \Delta_j^{\text{TPS}}\|_2}{\|u_i - u_j\|_2}$$

for neighbouring spots $i \sim j$, and let

$$G_i = \max_{j: j \sim i} g_{ij}.$$

The resulting values are robustly standardised,

$$Z_i^{\text{TPS}} = \frac{G_i - \text{median}(G)}{1.4826 \text{ median}_k |G_k - \text{median}(G)|}.$$

Large values indicate locations at which the globally smooth TPS fit requires an unusually rapid local change in displacement.

Binary H&E brightness gate. Let $S_{\text{HSV}}(z)$ and $V_{\text{HSV}}(z)$ denote the saturation and value channels of the H&E image, scaled to $[0, 1]$. We define the whiteness score

$$W(z) = V_{\text{HSV}}(z) \{1 - S_{\text{HSV}}(z)\}.$$

Hue is not used because it is unstable for nearly white pixels.

Vessels are segmented using Cellpose, StarDist or another validated segmentation procedure. Within each segmented vessel, the half of the pixels with the largest whiteness scores are retained as a vessel-lumen reference set $\mathcal{V}_{\text{lum}}$. The slide-specific threshold is

$$t_{\text{HE}} = \max \{0, Q_{0.05}(\{W(z) : z \in \mathcal{V}_{\text{lum}}\}) - 0.05\}.$$

The binary bright-pixel mask is

$$B_{\text{HE}}(z) = \mathbf{1} \{W(z) \geq t_{\text{HE}}\}.$$

The subtraction of 0.05 makes the criterion slightly more permissive than the observed vessel-lumen range, allowing smaller or partially stained tears to pass the H&E gate.

For MSI spot i , let $\mathcal{F}_i$ denote its approximate H&E footprint centred at $T_1(u_i)$. The spot-level gate is

$$B_{\text{HE},i} = \max_{z \in \mathcal{F}_i} B_{\text{HE}}(z).$$

Thus, the gate is activated whenever any part of the mapped MSI footprint overlaps a sufficiently bright H&E region.

Tear core and skeleton. The raw tear-core candidate set is

$$\mathcal{C}_0 = \{i : B_{\text{HE},i} = 1 \quad \text{and} \quad Z_i^{\text{TPS}} > z_{\text{cut}}\},$$

where z_{cut} is a method parameter with default value

$$z_{\text{cut}} = 4.$$

The H&E criterion is therefore only a permissive gate; the primary evidence for a tear is supplied by the displacement-gradient score.

A morphological closing with a one-MSI-pixel structuring element is applied to $\mathcal{C}_0$ to bridge isolated one-pixel gaps. The closed set is denoted by $\mathcal{C}$. It is skeletonised, isolated skeleton components are removed, and terminal branches shorter than two MSI pixel spacings are pruned. The resulting one-dimensional tear representation is denoted by

$$\Gamma.$$

Closing, skeletonisation and pruning are performed before any dilation of the tear mask.

Trusted, movable and core spots. Let $\mathcal{N} = \{1, \dots, n\}$ denote the valid MSI spots, and define

$$\mathcal{H} = \{i : Z_i^{\text{TPS}} > z_{\text{cut}}\} \cup \text{dilate}_1(\mathcal{C}).$$

The trusted and movable sets are

$$\mathcal{G} = \mathcal{N} \setminus \mathcal{H}$$

and

$$\mathcal{U} = \mathcal{H} \setminus \mathcal{C}.$$

Consequently,

$$\mathcal{N} = \mathcal{G} \dot{\cup} \mathcal{U} \dot{\cup} \mathcal{C}.$$

Spots in $\mathcal{G}$ remain fixed at their TPS-aligned positions. Spots in $\mathcal{U}$ are candidates for local refinement. Spots in $\mathcal{C}$ are not used to estimate the tear-aware deformation but receive coordinates by post-fit displacement extension.

Cut graph and anchored components. Let E be the eight-neighbour graph on the non-core spots $\mathcal{N} \setminus \mathcal{C}$. An edge is retained only when the straight line segment joining its endpoints does not intersect the tear skeleton:

$$E_\Gamma = \{(i, j) \in E : [u_i, u_j] \cap \Gamma = \emptyset\}.$$

This removes smoothness coupling across the tear while retaining ordinary coupling within each side.

Only connected components of $\mathcal{U}$ that remain connected through E_Γ to at least one trusted spot in $\mathcal{G}$ are adjusted. Their union is denoted by $\mathcal{U}^*$. Components without a trusted anchor are flagged as unresolved.

The fixed trusted boundary is

$$\partial\mathcal{U}^* = \{i \in \mathcal{G} : \exists j \in \mathcal{U}^* \text{ such that } (i, j) \in E_\Gamma\}.$$

Let

$$\mathcal{R} = \mathcal{U}^* \cup \partial\mathcal{U}^*.$$

On the graph induced by $\mathcal{R}$ and E_Γ , let

$$L_\Gamma = D_\Gamma - A_\Gamma$$

be the combinatorial Laplacian and

$$L_{\text{rw}, \Gamma} = D_\Gamma^{-1} L_\Gamma$$

its random-walk-normalised form.

Fixed-standardisation agreement term. The fixed MSI metafeature and the TPS-aligned H&E target are standardised once using the trusted set:

$$\tilde{q}_i = \frac{q_i^{(0)} - \bar{q}_\mathcal{G}}{\text{sd}_\mathcal{G}(q^{(0)})}$$

and

$$\tilde{s}_T(i) = \frac{s_T(i) - \bar{s}_\mathcal{G}^{(1)}}{\text{sd}_\mathcal{G}(s_{T_1})},$$

where

$$\bar{s}_\mathcal{G}^{(1)} = \frac{1}{|\mathcal{G}|} \sum_{i \in \mathcal{G}} s_{T_1}(i).$$

The centring and scaling constants remain fixed while T changes.

For a movable set A , define

$$Q_A(T) = \frac{1}{|A|} \sum_{i \in A} \tilde{q}_i \tilde{s}_T(i).$$

This fixed-standardisation cross-covariance avoids estimating noisy local means and variances while rewarding aggregate agreement between the MSI metafeature and the H&E target.

Local tear-aware refinement. For a residual displacement field Δ on $\mathcal{R}$, with

$$\Delta_i = 0, \quad i \in \partial\mathcal{U}^*,$$

define

$$R(\Delta) = \frac{1}{|\mathcal{U}^*|} \|(L_{\text{rw},\Gamma}\Delta)_{\mathcal{U}^*}\|_F^2.$$

The Laplacian is applied separately to the two displacement coordinates, which are expressed in MSI-pixel units. The local tear-aware displacement is estimated as

$$\hat{\Delta} = \arg \max_{\Delta} [Q_{\mathcal{U}^*}(T_1 + \Delta) - \lambda_{\text{local}} R(\Delta)],$$

subject to

$$\Delta_i = 0, \quad i \in \partial\mathcal{U}^*.$$

No hard displacement bound is imposed. The squared-Laplacian penalty suppresses pixel-scale oscillations and local curvature, whereas removal of skeleton-crossing edges permits relative movement across a detected tear.

Calibration of the local regularisation strength. We write

$$\lambda_{\text{local}} = 2^k.$$

Because the agreement and regularisation terms are both averaged over movable spots, and the latter uses the random-walk-normalised Laplacian, $k = 0$ provides a meaningful central starting value.

Let

$$\mathcal{U}_1, \dots, \mathcal{U}_M$$

be the connected components of $\mathcal{U}^*$, with sizes $m_j = |\mathcal{U}_j|$. Ten target control-region sizes are chosen from equally spaced empirical quantiles of $m_1, \dots, m_M$. If fewer than ten components are present, the observed sizes are reused until ten target sizes have been assigned.

Candidate seeds are taken from the one-pixel morphological erosion of $\mathcal{G}$. The first seed is sampled uniformly. Subsequent seeds are drawn according to the k -means++ rule, with probability proportional to

$$d(v, \mathcal{S})^2,$$

where $\mathcal{S}$ denotes the previously selected seeds and $d(v, \mathcal{S})$ is the Euclidean distance to the nearest such seed. This favours spatially distributed control regions.

For a seed with target size m , the corresponding pseudo-movable region consists of the m vertices having the smallest graph distance to the seed in the eight-neighbour graph restricted to $\mathcal{G}$. The region is accepted only if its complete one-pixel outer boundary also lies in $\mathcal{G}$. The same pseudo-regions are used for all tested values of k .

For each pseudo-region, its interior is allowed to move while its trusted boundary remains fixed. Let

$$M(k) = Q_{0.95} \left(\left\{ \left\| \hat{\Delta}_i^{(k)} \right\|_2 : i \text{ belongs to a pseudo-movable region} \right\} \right).$$

The calibration criterion is

$$M(k) \leq 0.5$$

MSI pixels.

The search starts at $k = 0$. If the criterion is satisfied, k is decreased in steps of one until it first fails, and the preceding passing value is selected. If the criterion fails at $k = 0$, k is increased in steps of one until it first holds. Thus, the selected value is the weakest regularisation strength on the power-of-two grid that prevents appreciable noise-driven movement in trusted tissue. All candidate fits are initialised at zero displacement.

Harmonic extension into the tear core. Core spots neither contribute to the cross-modal agreement term nor participate in the regularisation problem used to estimate the surrounding non-core deformation. This prevents uncertain or damaged core measurements from altering the inferred alignment of intact tissue.

After fitting the non-core displacement, each connected core region is assigned to the closest connected component of the cut graph. Ambiguous assignments are flagged as unresolved. Let E_{Γ}^{ext} contain graph edges joining assigned core spots to one another and to the corresponding fitted non-core component. The core displacements are obtained from

$$\hat{\Delta}_{\mathcal{C}} = \arg \min_{\Delta_{\mathcal{C}}} \sum_{(i,j) \in E_{\Gamma}^{\text{ext}}} \|\Delta_i - \Delta_j\|_2^2,$$

while all fitted non-core displacements remain fixed. Thus, coordinates are estimated for resolved core spots by harmonic extension, but these spots neither influence nor receive any reward from the registration objective.

Output labels and numerical warning. Each MSI spot is labelled as

$$\begin{aligned} \text{trusted,} & & i \in \mathcal{G}, \\ \text{locally adjusted,} & & i \in \mathcal{U}^*, \\ \text{tear-core extrapolated,} & & i \in \mathcal{C} \text{ with an unambiguous extension,} \\ \text{unresolved,} & & \text{otherwise.} \end{aligned}$$

A numerical warning is issued whenever

$$\max_i \|\hat{\Delta}_i\|_2 \geq 15$$

MSI pixels.

Post-alignment diagnostic metafeature. After estimating the final transformation $\hat{T}$, the smooth anatomical target is evaluated at all resolved non-core MSI spots,

$$s_{\hat{T}}(i) = f(\hat{T}(u_i)).$$

Using these spots, the diagnostic ridge coefficients are estimated from

$$\left(\hat{\alpha}^{(\text{final})}, \hat{\beta}^{(\text{final})}\right) = \arg \min_{\alpha, \beta} \left\{ \frac{1}{n_{\text{resolved}}} \|s_{\hat{T}} - \alpha \mathbf{1} - \tilde{X} \beta\|_2^2 + 0.05 \|\beta\|_2^2 \right\}.$$

After normalisation and sign orientation,

$$\hat{w}^{(\text{final})} = \frac{\hat{\beta}^{(\text{final})}}{\|\hat{\beta}^{(\text{final})}\|_2}.$$

Metafeature-pattern stability is summarised by

$$\rho_{\text{score}} = \left| \rho \left(\tilde{X} \hat{w}^{(0)}, \tilde{X} \hat{w}^{(\text{final})} \right) \right|.$$

References

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